Omar Ehab Elgedawy

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Education

Arab Academy for Science, Technology and Maritime Transport (AAST) Alamein, Egypt
Full Scholarship, BSc in Artificial Intelligence *2024 – 2028*
Coursework: Calculus, Linear Algebra, Probability and Statistics, Discrete Mathematics, Fundamental of Datascience, Computer Organization, Data Structures & Algorithms, Database, Networking, Operating Systems, Machine Learning.

Experience

Nile University, Research Intern – Quantum Finance Group Jul 2026 – Present
Cairo, Egypt · Hybrid

- Implementing a **quantum machine learning** feature-engineering pipeline for market-direction prediction, based on a peer-reviewed **quantum finance** research paper, combining quantum feature maps with classical **ML** models.
- Reproducing and extending published methodology; building a backtesting framework to benchmark quantum-enhanced features against classical baselines, with source code published on **GitHub**.

Auvnet, C++ Software Engineer (Computer Vision, Contract) Dec 2025 – May 2026
Dubai, UAE · Remote

- Architected a modular **C++** engine using **OpenCV** and **MediaPipe** via a plug-and-play OOP architecture.
- Optimized realtime performance to **40+ FPS** across **6+ layers** via memory management and algorithmic tuning.
- Engineered **14 filters** by mapping **468 landmarks** to custom HSV/RGB and texture-warping algorithms.

Coach Academy, Instructor and Coach Jan 2025 – Dec 2025
Cairo, Egypt · Remote

- Taught **40+ students** competitive programming, algorithms, and problem-solving through hands-on coaching.

NEAR Protocol, Freelance Competitive Programming Specialist Sep 2022 – Oct 2024
San Francisco, CA, USA · Online Freelancing

- Solved and documented **120+ competitive programming problems** in C++ to support AI model training.
- Authored **40+ LaTeX tutorials** on high-level DSA topics like tree and graph, DP algorithms, and core CS topics.
- Reviewed and debugged **60+ failed Codeforces solutions** by analyzing errors and delivering fixes.

Projects

Autonomous Service Robot (RoboCup @Home), Computer Vision & Tracking System Apr 2026

- Developed the computer vision and tracking system for an autonomous service robot using **YOLO**.
- Implemented real-time person detection and spatial tracking to autonomously locate and interact with guests in unknown room environments using **ROS**, **YOLO**, and **Python**.

Windows Shell (C++), Windows-Compatible Shell Implementation Aug 2025

- Engineered a custom Windows shell in C++ with readline, advanced editing, and persistent command history.
- Implemented built-in commands, tab completion, and robust I/O redirection leveraging **Windows APIs**.
- Enhanced usability with arrow key navigation, history management, and Windows-specific path handling.

Digital Clock with AM/PM Indicator, Simulation and Hardware Implementation May 2025

- Constructed a 12-hour digital clock with AM/PM indicator using D flip-flops and 555 timer in **Proteus**.
- Prototyped the circuit on breadboard with 7490 counters, 7447 decoders, and 7-segment displays.
- Implemented reset detection at 12 hours with AM/PM toggling logic for full cycle completion.

Volunteering

AAST ICPC Club, Founder, President, & Training Head Sep 2024 – Present

- Founded AAST's ICPC Club and managed PR, Media, HR, and Training teams.
- Designed a **3-level curriculum** from C++ basics to advanced algorithms; led weekly training camps.
- Mentored **300+ students**, qualifying **4 teams to ECPC** and **1 team to ACPC**.

FCDS ICPC Community, Alexandria University, Instructor Oct 2024 – May 2025

- Taught competitive programming at the Faculty of Computers and Data Science (FCDS).

Achievements

Codeforces — Expert (Top 5% globally), solved 2500+ problems across various online judges
RoboCup @Home — National Champion (1st Place in Egypt) & International Qualifier (2026)
African & Arab Collegiate Programming Contest (ACPC) — 5× Qualifier, 6th Place (2024)
Egyptian Olympiad in Informatics (EOI) — 2× Bronze Medalist (2020, 2023)
International Informatics Olympiad Teams (IIOT) — National Champion, Finalist (2022)
Egyptian Collegiate Programming Contest (ECPC Teens) — Silver Medal (2021)

Skills

C/C++ · Python · Data Structures · Algorithms · Problem Solving · OOP · OpenCV · Mediapipe · SQL · Pandas · Matplotlib · Seaborn · Digital Circuits · Logic Design · Leadership · Mentorship · Team Management · ROS · YOLO · Qiskit · Quantum Computing · Quantum Machine Learning